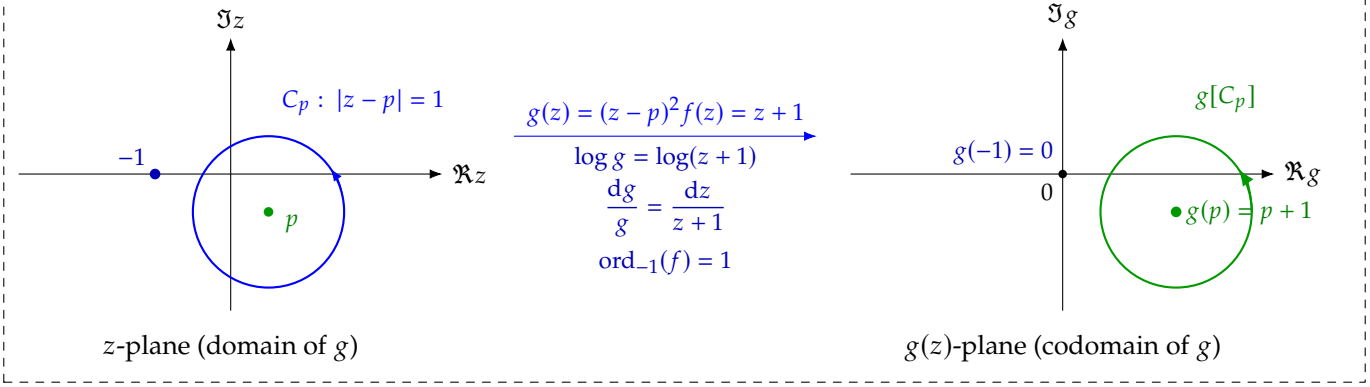
$\Im z$
$C_p : |z - p| = 1$
-1
$\Re z$
p
$g(z) = (z - p)^2 f(z) = z + 1$
$\log g = \log(z + 1)$
$\frac{dg}{g} = \frac{dz}{z + 1}$
$\mathrm{ord}_{-1}(f) = 1$
$\Im g$
$g[C_p]$
$g(-1) = 0$
0
$\Re g$
$g(p) = p + 1$
z-plane (domain of g)
$g(z)$-plane (codomain of g)